Paralog dS landscape of 7,384 fungal genomes (BFD): identical-copy excess and recent duplication signals

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Summary

This report describes node-level synonymous divergence (dS) of paralogs in 7,384 genomes with a finished paralogoscope table. The method is: wgd dmd families, identical-CDS collapse, yn00 dS on sampled pairs (k = 5 per node, gates dS <= 2.5 and >= 30 codons), and nodes from a neighbour-joining tree on protein identity. All values below are computed from `tables/` in this folder. Nothing here is a claim about individual genomes without independent evidence. A flag from dS is a screening result.

Limits that apply to every result:

- yn00 dS reads low above about 0.5 and saturates near 2 (Section 8). Only dS 0 to about 1.5 carries position information. The median share of nodes with a usable dS per genome is 0.26. The rest lie beyond the identity window.
- The data cannot detect old fungal WGDs (Saccharomycetaceae, Mucorales). They fall in the old or saturated mass. Question 2 is therefore: which genomes carry a recent genome-scale duplication (hybrid, heterokaryon or dikaryon assembled with both genomes, or a mixed assembly).
- Aneuploidy is not visible.
- Clade differences are partly confounded with assembly source and technology (Section 6).

1. Data and bins

Bins of node dS: **Z** identical (dS = 0), **H** (0, 0.05], **Y** (0.05, 0.3], **MID** (0.3, 1.5], **OLD** (1.5, 2.5], and **BEYOND** (no usable dS). Shares use two denominators: all nodes (`shareAll_*`) and nodes with a usable dS. Genome-level intervals come from a family-level cluster bootstrap. Genome statistics are medians per clade, and genomes are never pooled by node.

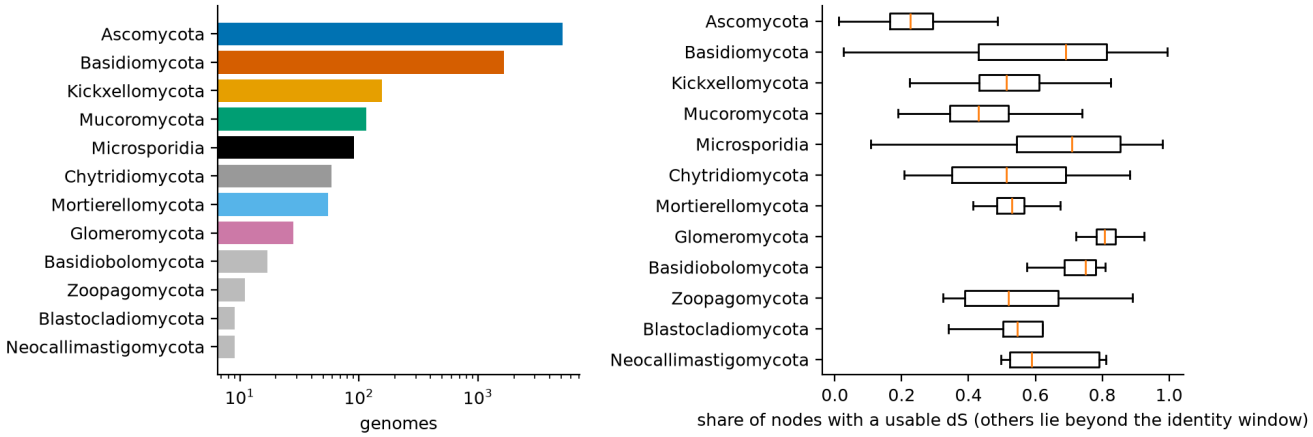


Figure 1: Share of nodes with a usable dS, by phylum.

Share of nodes with a usable dS, by phylum.

2. Composition by taxonomic group

Phylum medians over genomes (columns `med_*`; bootstrap intervals are in `tables/clade_summary_phylum.tsv`)

phylum	n_genomes	n_genera	med_frac_nodes_shareAll_sH	med_shareAll_sH	med_shareAll_sY	med_shareAll_sMID	med_shareAll_sOLD	med_shareAll_sBEYOND	med_nodes_per_sh	med_dS_geomean	med_dS_sd_log	med_dS_geomean_lo95
Ascomycota	5160	1215	0.225	0.000287	0.00099	0.00733	0.0354	284	0.518	0.4	0.0972	
Basidiomycota	1654	476	0.69	0.00378	0.0178	0.142	0.234	446	0.471	0.7	0.288	
Kickxellomycota	157	15	0.514	0.00122	0.0126	0.107	0.143	257	0.552	0.8	0.181	
Mucoromycota	116	39	0.43	0.00731	0.0103	0.0464	0.113	473	0.508	11.1	1.34	
Microsporidia	91	48	0.709	0.0864	0.0788	0.17	0.167	292	0.567	0.3	0.806	
Chytridiomycota	59	39	0.514	0.00664	0.00816	0.0409	0.122	360	0.537	2.4	0.328	
Mortierellomycota	55	14	0.53	0.00322	0.00102	0.0373	0.163	427	0.578	7.5	0.986	
Glomeromycota	28	14	0.807	0.00196	0.00727	0.186	0.428	654	0.377	1	3.02	
Zoopagomycota	11	8	0.519	0.0236	0.00918	0.0492	0.144	315	0.563	1.9	0.155	

Fifteen largest classes (n >= 10 eligible):

class	n_genomes	n_genera	med_shareAllZ	AllZ_shareAllH	AllH_shareAllY	AllY_shareAllMID	AllMID_nodes_per_1000_genes	med_shared_genera_fa
Sordariomycetes	1614	336	0.00018	0.000482	0.00296	0.0253	315	0.499
Agaricomycetes	1172	350	0.00483	0.0281	0.188	0.301	485	0.448
Eurotiomycetes	744	92	0.000181	0.000796	0.00584	0.0312	354	0.479
Pichiomycetes	665	87	0	0.00164	0.0154	0.0714	225	0.584
Dothideomycetes	613	235	0.000311	0.000604	0.00299	0.0216	291	0.507
Lecanoromycetes	408	176	0	0.000309	0.00299	0.0361	262	0.526
Saccharomycetes	365	52	0.001	0.00395	0.0239	0.0752	243	0.596
Leotiomycetes	236	95	0.000723	0.00157	0.00918	0.0409	312	0.489
Tremellomycetes	220	42	0.000714	0.00153	0.00764	0.0305	230	0.538
Dipodascomycetes	210	27	0.000604	0.0016	0.0192	0.078	233	0.564
Kickxellomycetes	141	7	0.00109	0.0113	0.105	0.143	251	0.551
Pezizomycetes	121	32	0.00398	0.00743	0.0912	0.153	306	0.525
Mucoromycetes	106	36	0.00793	0.0103	0.0488	0.116	479	0.504
Microsporidia (unassigned)	91	48	0.0864	0.0788	0.17	0.167	292	0.567
Microbotryomycetes	77	18	0.00342	0.0038	0.0235	0.0848	251	0.54

Node-dS distributions by class. Grey band: yn00 saturation zone (no position claim). Black bar: identical-copy share.

Bin shares by order (genomes / genera in labels).

Nodes per 1,000 genes against dS-bin mass.

3. How much of the variation is taxonomic

Nested random-effects model (taxonomy as a phylogeny proxy; branch lengths ignored), with and without assembly covariates.

response	adjustment	phylum	class	order	family	genus	genome (residual)
BEYOND	no covariates	0.24	0.31	0.038	0.063	0.11	0.24
BEYOND	with covariates	0.34	0.23	0.04	0.063	0.11	0.22
H	no covariates	0.22	0.17	0.0062	0.11	0.12	0.38
H	with covariates	0.39	0.079	0.018	0.055	0.092	0.36
MID	no covariates	0.2	0.3	0.081	0.068	0.11	0.23
MID	with covariates	0.13	0.29	0.09	0.067	0.13	0.29
Y	no covariates	0.16	0.22	0.041	0.085	0.12	0.37
Y	with covariates	0.098	0.24	0.053	0.072	0.11	0.43
identical	no covariates	0.18	0.16	0.01	0.12	0.077	0.45
identical	with covariates	0.36	0.061	0.02	0.069	0.069	0.42

response	adjustment	phylum	class	order	family	genus	genome (residual)
nodes_per_1000	no covariates	0.31	0.32	0.019	0.078	0.094	0.18
nodes_per_1000	with covariates	0.3	0.23	0.018	0.086	0.17	0.19
paralog_frac	no covariates	0.26	0.23	0.014	0.055	0.096	0.35
paralog_frac	with covariates	0.42	0.14	0.0061	0.053	0.15	0.23
two_gene_fam	no covariates	0.16	0.16	0.02	0.052	0.1	0.5
two_gene_fam	with covariates	0.15	0.35	0.022	0.014	0.097	0.37

Reading: share is the fraction of total random variance at that level. A large residual or genus share means the trait varies within clades. Large phylum to order shares mean clade identity predicts it.

Variance components by taxonomic level.

4. Question 1: excess of dS = 0 and near-zero paralogs

Outlier flags use robust z within clade (order, then class, then phylum; n >= 10; MAD floors) on unshrunken values. Flagged rows: identical excess 594, H excess 18, Y excess 49, MID excess 111, duplication-dense 102, 2-gene-family excess 210 (rows in `outlier_flags.tsv`: 833). These counts are screening hits, not confirmed events. No multiplicity-adjusted claim is made per genome.

Identical-copy excess split by simple descriptors (BUSCO duplicated % and largest identical group). The two groups are a heuristic, not a diagnosis:

profile	n	med_busco_dup	med_largest_group	med_te_per_1000	med_ident_per_1000
TE/multicopy-like (large group, low BUSCO dup)	248	0.7	9	0.127	11.8
assembly-duplication-like (BUSCO dup >= 20%)	61	53.7	6	0.283	39.8
other	285	1.1	3	0.13	10

Classes with the most identical-excess flags:

phylum	class	n_flagged
Ascomycota	Sordariomycetes	145
Ascomycota	Eurotiomycetes	80
Ascomycota	Dothideomycetes	71
Basidiomycota	Agaricomycetes	65
Ascomycota	Pichiomycetes	62
Ascomycota	Saccharomycetes	41
Ascomycota	Leotiomycetes	39
Ascomycota	Lecanoromycetes	20

phylum	class	n_flagged
Basidiomycota	Tremellomycetes	19
Basidiomycota	Microbotryomycetes	11
Ascomycota	Dipodascomycetes	10
Kickxellomycota	Kickxellomycetes	9

Interpretation (from the framing review, `fable_context.md`): a flag with high BUSCO duplication and many small groups points to duplicated contigs or haplotigs. A flag with low BUSCO duplication, large groups and high TE counts points to TE-derived models or multicopy arrays. H excess without Z points to haplotype or heterozygosity depth. Tests that would separate these are listed in Section 10.

Identical-copy statistics by clade.

Young-node mass (H, Y) against BUSCO duplication and N50.

5. Question 2: dS peaks suggesting recent genome-scale duplication, hybrid or mixed assembly

Peaks come from a Gaussian mixture on log dS (dS > 0). A peak counts as *stable* only if all of these hold: node bootstrap >= 0.8, present at >= 100 codons, stable under pair resampling (>= 0.8), and position in 0.05 to 1.5. Components outside that window are not treated as duplication peaks.

The catalogue file lists 1181 components in 1132 genomes. Of these, 1181 pass the stability rules above (the catalogue is already filtered; this is a check). Stable and with assembly support (BUSCO dup >= 40% and gene count >= 1.5x clade median): 115 components in 113 genomes.

phylum	components	genomes
Basidiomycota	668	636
Ascomycota	337	328
Kickxellomycota	52	52
Microsporidia	39	36
Glomeromycota	28	25
Chytridiomycota	17	17
Mucoromycota	12	12
Basidiobolomycota	13	11
Entomophthoromycota	5	5
Neocallimastigomycota	3	3
Zoopagomycota	2	2
Mortierellomycota	2	2
Blastocladiomycota	1	1
Monoblepharomycota	1	1
unassigned	1	1

Top 40 stable peaks with assembly support (sorted by BUSCO duplicated %):

genome	phylum	class	peak_dS	support_node	busco_duplicated	genes_ratio	shared_2gene_fam
Epichloe_coenophiala_ATCC_90664	Ascomycota	Sordariomycetes	0.0659	1	91.5	1.99	0.186
Scleroderma_bermudense	Basidiomycota	Agaricomycetes	1.48	0.97	89.9	1.72	0.788
Scleroderma_bermudense	Basidiomycota	Agaricomycetes	0.511	1	89.9	1.72	0.788
Puccinia_polysora_GD1018	Basidiomycota	Pucciniomycetes	0.13	0.94	87.9	2.25	0.766

genome	phylum	class	peak_dS	support_node	busco_duplicated	gene_ratio_shared	gene_fam
Favolaschia_claudopus_BRFM_2984	Basidiomycota	Agaricomycetes	0.152	1	87.5	4.57	0.136
Lactarius_tabidus_BRFM_1826	Basidiomycota	Agaricomycetes	0.594	1	86.2	2.37	0.684
Coniochaeta_sp._2T2.1_PT2906	Ascomycota	Coniochaetales	0.278	1	83.5	2.26	0.813
Rhizopus_sp._PT2906_PT2906	Mucoromycota	Mucoromycetes	0.0827	1	82	1.71	0.709
Agroathelia_delphinioides_SD	Basidiomycota	Agaricomycetes	0.278	1	81.3	1.55	0.781
Exophiala_dermatitidis_EXF-5586	Ascomycota	Eurotiomycetes	0.191	1	80.8	1.54	0.86
Cryoendolithus_antarcticus_CCFEE_5527-51	Ascomycota	Dothideomycetes	0.551	1	79.9	1.65	0.845
Friedmanniomyces_endolithicus_CCFEE_6310	Ascomycota	Dothideomycetes	0.687	1	79.2	1.52	0.853
Vermiconidia_calcicola_CCFEE_5714	Ascomycota	Dothideomycetes	0.258	1	78.8	1.86	0.83
Ogataea_polymorpha_Ogataea_polymorpha_CBS_1977	Ascomycota	Pichiomycetales	0.164	1	78.6	2.92	0.32
Aureobasidium_sp._EXF-3400	Ascomycota	Dothideomycetes	0.805	1	77.6	2.12	0.815
Aureobasidium_sp._EXF-12298_EXF-12298	Ascomycota	Dothideomycetes	0.805	1	77.6	2.05	0.823
Verticillium_longisporum_BDS89	Ascomycota	Sordariomycetes	0.13	1	77.2	1.54	0.791
Aureobasidium_sp._DW-1_DW-1	Ascomycota	Dothideomycetes	0.164	1	76.8	2.08	0.688
Neonothopanus_nambi_BW_2379	Basidiomycota	Agaricomycetes	0.0566	1	76.1	2.32	0.598
Agroathelia_rolfsii_MR10	Basidiomycota	Agaricomycetes	0.177	1	75.7	1.61	0.72
Apiotrichum_sp._PDH-33048_PDH-33048	Basidiomycota	Tremellomycetes	0.152	1	75	1.86	0.878
Apiotrichum_sp._PA4207_PA4207	Basidiomycota	Tremellomycetes	0.152	1	74.9	1.87	0.877
Apiotrichum_sp._Y4204_Y4204	Basidiomycota	Tremellomycetes	0.152	1	74.9	1.86	0.877
Apiotrichum_sp._Y4201_Y4201-1	Basidiomycota	Tremellomycetes	0.152	1	74.7	1.85	0.878
Apiotrichum_sp._PDH-33066_PDH-33066	Basidiomycota	Tremellomycetes	0.152	1	74.2	1.86	0.876
Apiotrichum_sp._PDH-33098_PDH-33098	Basidiomycota	Tremellomycetes	0.152	1	74.2	1.86	0.879
Debaryomyces_hansenii_PMW_3.1.1880	Ascomycota	Pichiomycetales	0.278	1	73.8	2.16	0.861
Pisolithus_thermaeus_11	Basidiomycota	Agaricomycetes	0.104	1	73.6	1.77	0.706
Hansenula_lynferdii_NRRL_Y-7723	Ascomycota	Saccharomycetes	0.85	1	73.5	2.11	0.861
Pisolithus_croceorrhizus_2A72	Basidiomycota	Agaricomycetes	0.104	1	73	1.95	0.724
Debaryomyces_sp._S3305_S3305	Ascomycota	Pichiomycetales	0.278	1	72.5	2.11	0.865
Aureobasidium_sp._EXF-12344_EXF-12344	Ascomycota	Dothideomycetes	0.805	1	71.2	2.02	0.807
Millerozyma_farinosa_KCTC_27753	Ascomycota	Pichiomycetales	0.324	1	71	1.91	0.885
Lodderomyces_elongisporus_RS	Ascomycota	Pichiomycetales	0.152	1	70.9	2.23	0.824
Neocatenulostroma_macrosporum_L3-BK-P2	Ascomycota	Dothideomycetes	0.324	1	69.9	2.33	0.783
Saccharomycopsis_fibuligera_Saccharomycopsis_cf._fibuligera_KJJ81	Ascomycota	Saccharomycetes	0.278	1	68	2.11	0.875
Rhodosporidiobolus_fluvialis_Rhodosporidiobolus_fluvialis_CBS_7579	Basidiomycota	Microbotryomycetes	1.09	1	68	2.48	0.767
Saccharomyces_cerevisiae_x_Saccharomyces_eubayanus_x_Saccharomyces_kudriavzevii_x_Saccharomyces_uvarum_Saccharomyces_cerevisiae_x_Saccharomyces_eubayanus_x_Saccharomyces_kudriavzevii_x_Saccharomyces_uvarum_CBS_12357	Ascomycota	Saccharomycetes	0.837	1	67.7	2.69	0.68
Lyophyllum_atratum_CBS_414.62	Basidiomycota	Agaricomycetes	0.104	1	67.4	1.55	0.707
Saccharomyces_sp._CEN.Y65_CEN.Y65	Ascomycota	Saccharomycetes	0.0827	1	67.4	1.84	0.9

Peak positions and masses of the stable catalogue.

5.1 Controls

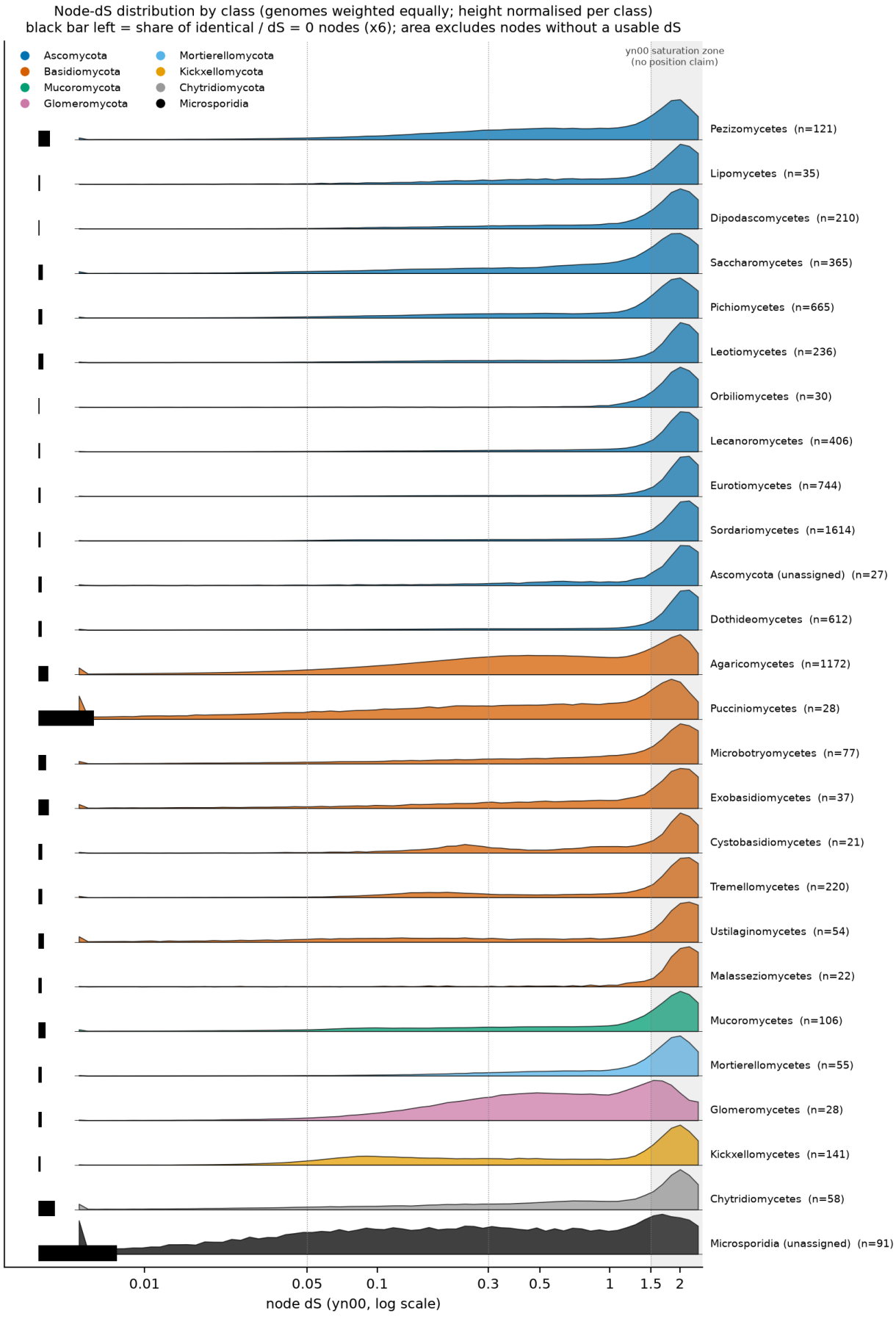


Figure 2: Node-dS distributions by class. Grey band: yn00 saturation zone (no position claim). Black bar: identical-copy share.

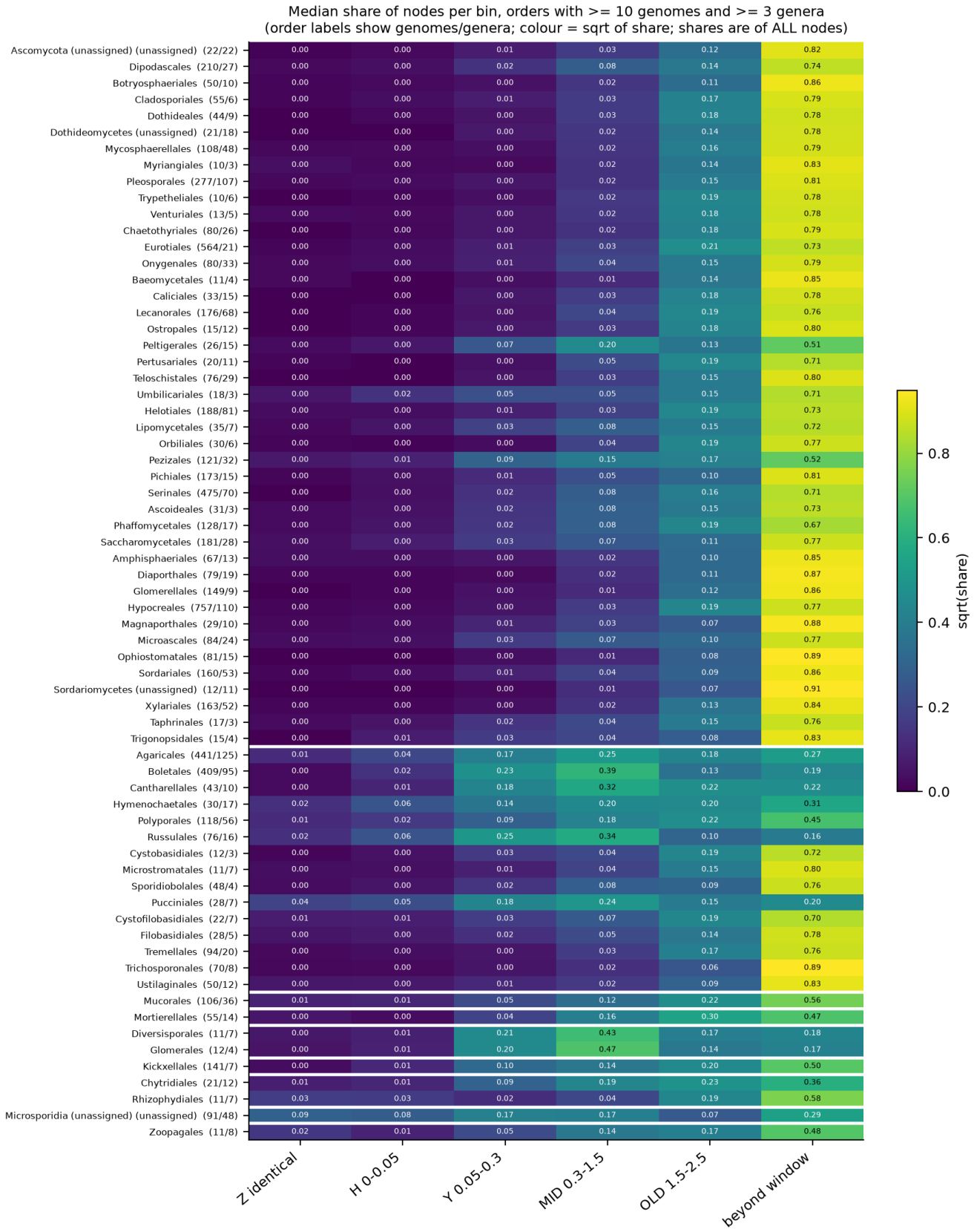


Figure 3: Bin shares by order (genomes / genera in labels).

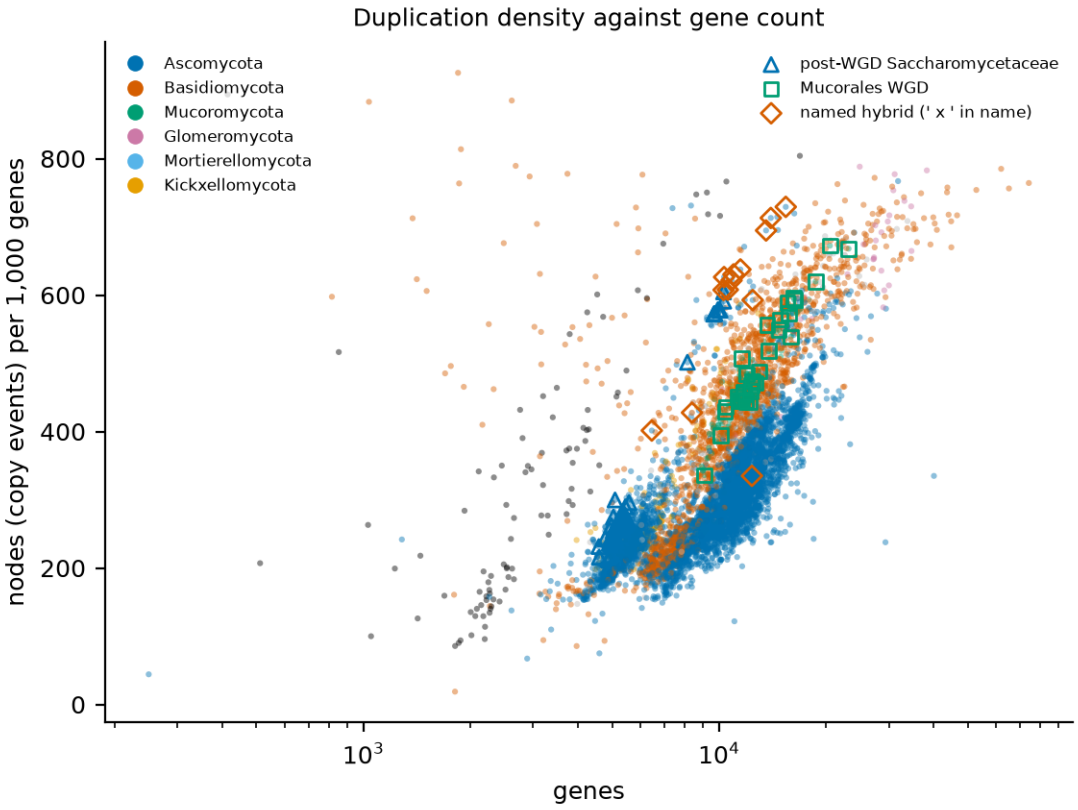


Figure 4: Nodes per 1,000 genes against dS-bin mass.

control_group	n	med_Z	med_H	med_Y	med_2gene	med_busco	med_gene_ratio	with_supported_
Mucorales WGD	32	0.00894	0.0186	0.0623	0.518	12.2	1.03	0.25
literature hybrid species (strain may be collapsed)	12	0.00123	0.00381	0.0308	0.691	57.3	1.65	0.5
named hybrid (' x ' in name)	13	0.00263	0.00723	0.0311	0.674	61.7	2.11	0.923
post-WGD Saccharomycetaceae	53	0.0021	0.00769	0.0375	0.66	3.7	1.02	0.226
pre-WGD Saccharomycetaceae	64	0.00106	0.00226	0.0145	0.609	0.9	0.956	0.172

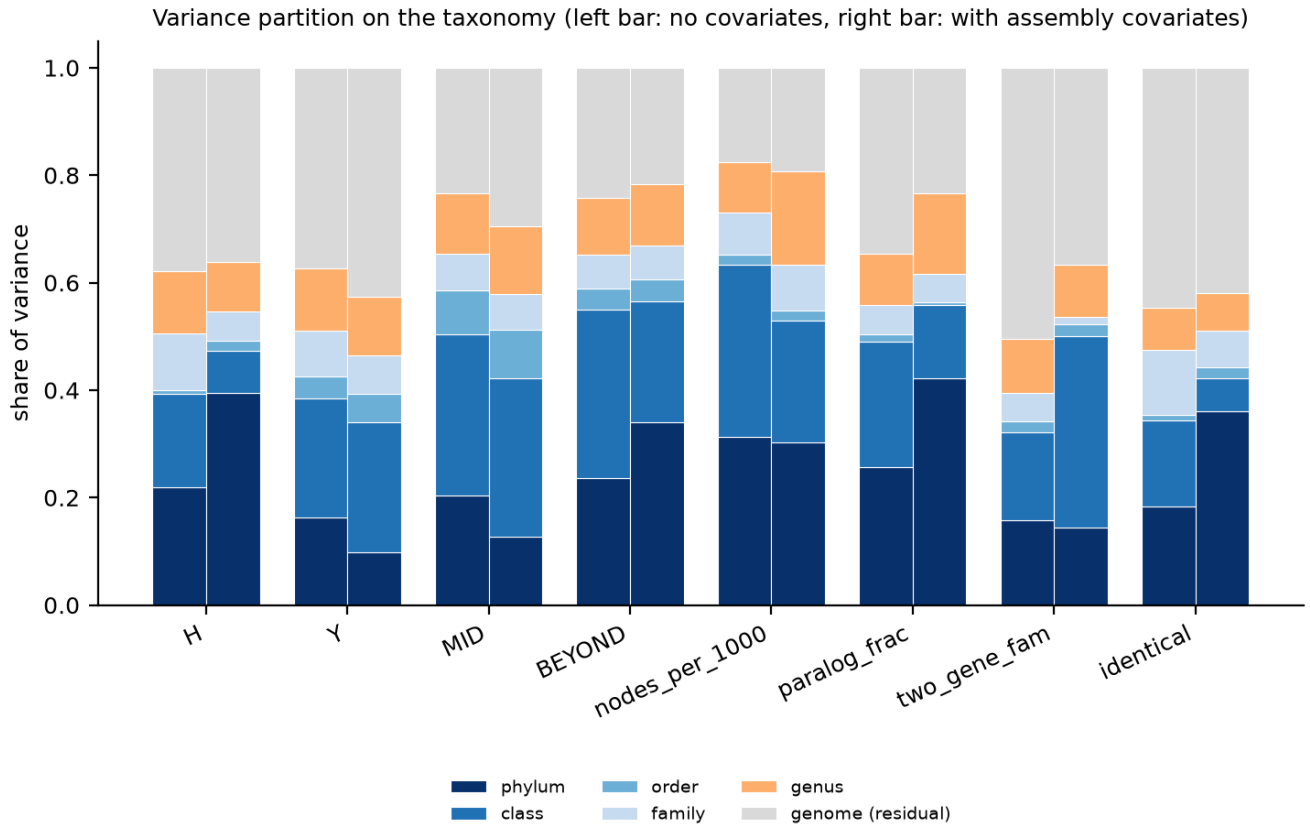


Figure 5: Variance components by taxonomic level.

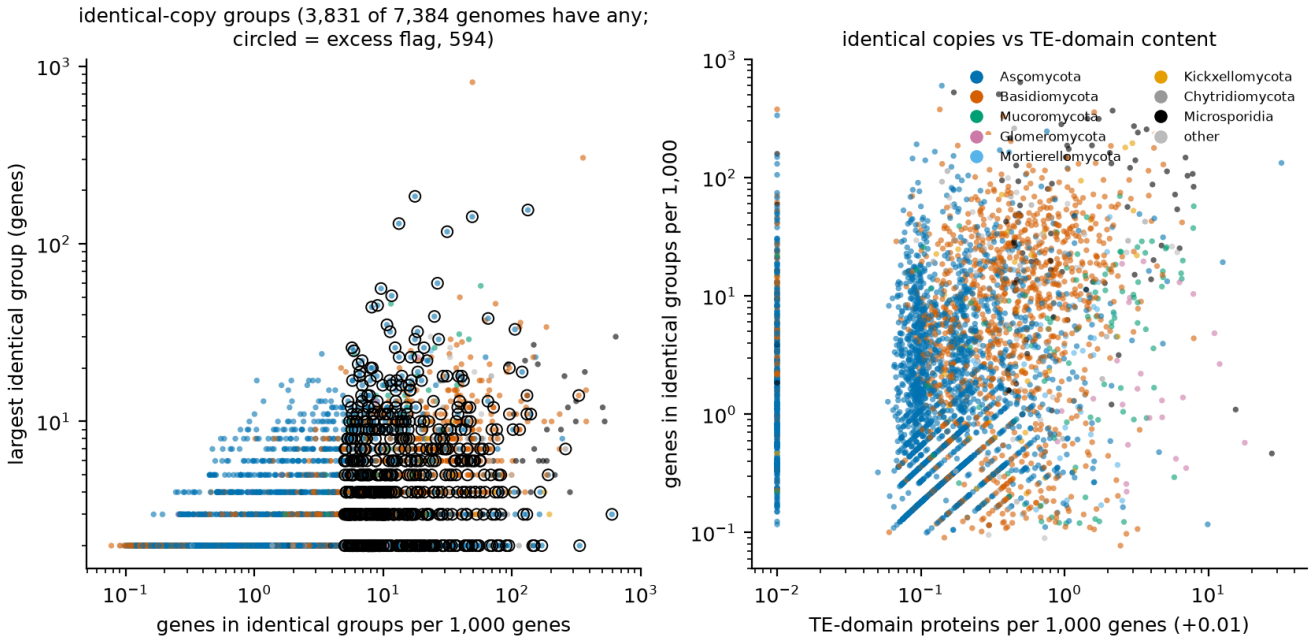


Figure 6: Identical-copy statistics by clade.

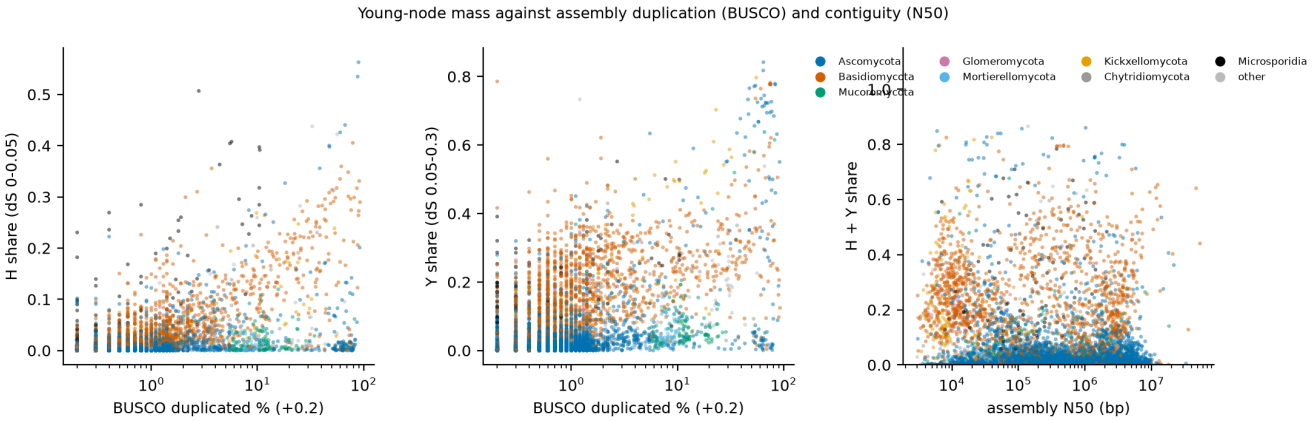


Figure 7: Young-node mass (H, Y) against BUSCO duplication and N50.

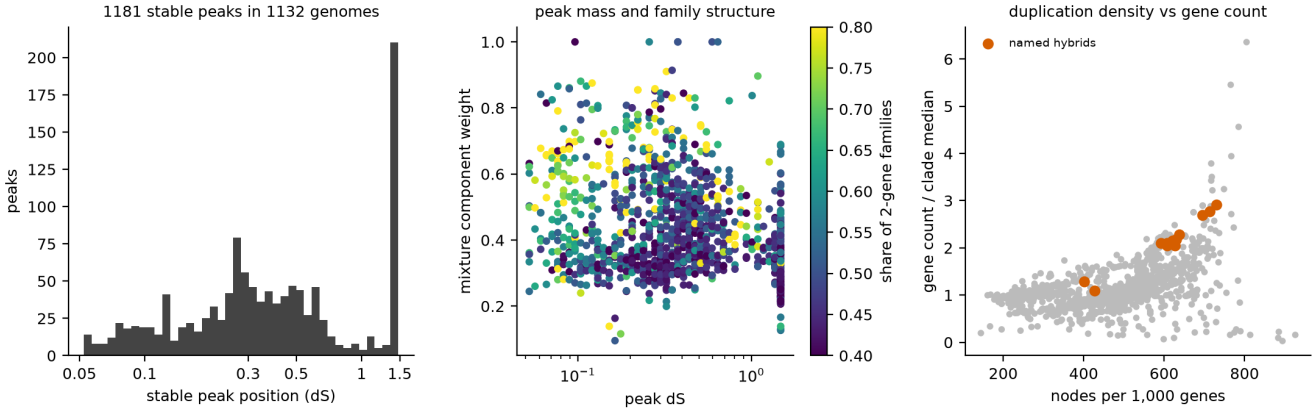


Figure 8: Peak positions and masses of the stable catalogue.

Named hybrids (’ x ’ in name) and literature hybrids are the positive controls for a mid-dS peak. Collapsed assemblies of hybrids show no peak. Saccharomycetaceae and Mucorales are controls for old mass and 2-gene structure, not for a peak, because their WGD is older than the dS window. Several named controls show sharp peaks near dS 0.2 to 0.3 or near 1.3 to 1.5 in the figure. A peak at about 1.3 to 1.5 lies near the yn00 saturation zone, so its position is a lower bound on the true divergence.

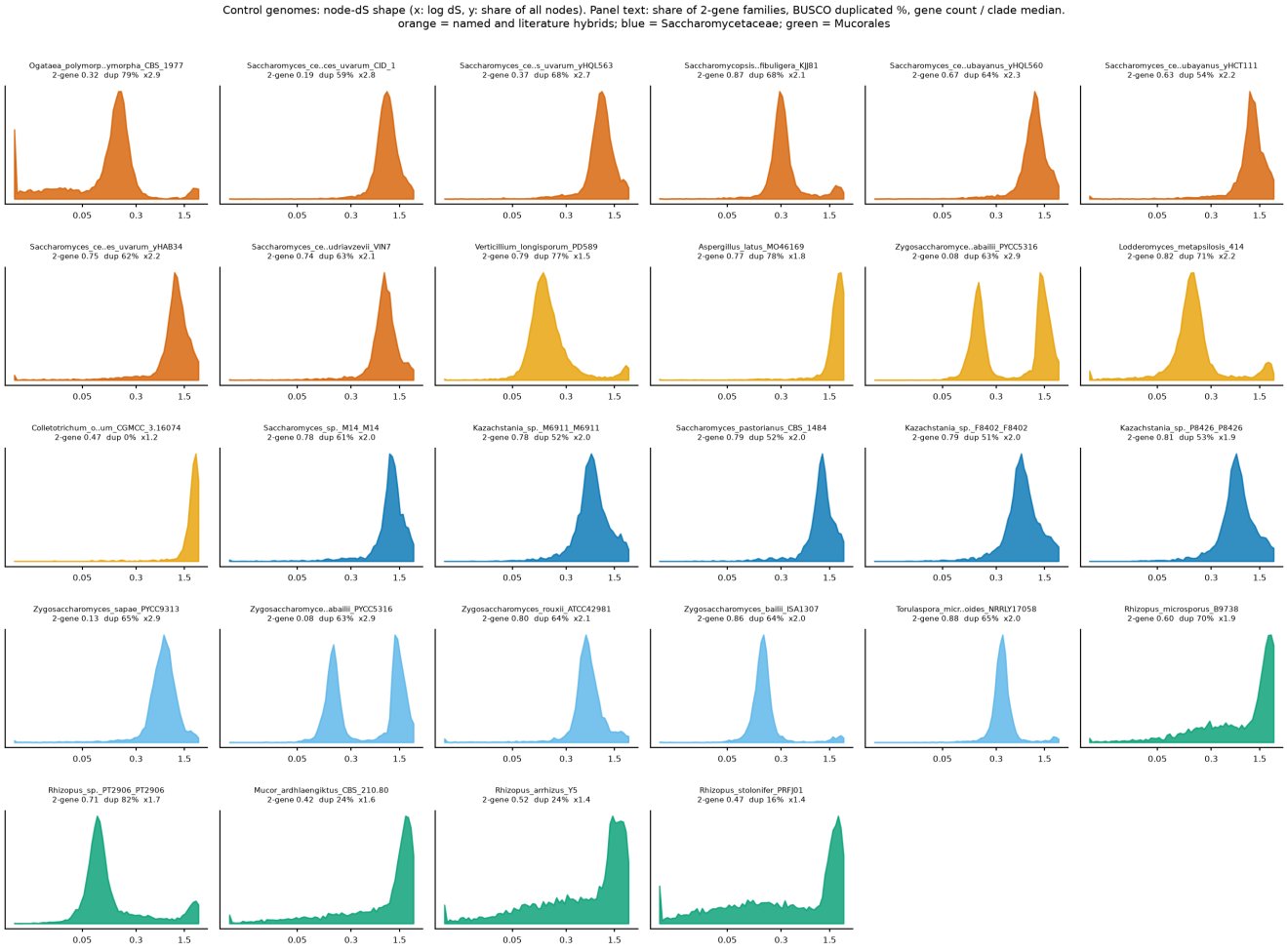


Figure 9: Control genomes: node-dS shape.

Control genomes: node-dS shape.

5.2 Interior peaks (centre <= 1.0) detected from histogram shape, cross-referenced with BUSCO

This screen uses the per-genome log-dS histogram directly (70 bins, 0.005 to 2.5), not the mixture model. A peak is a local maximum of the 3-bin smoothed density with prominence >= 35% of its height, centre between 0.03 and 1.0, and at least 8% of the genome’s tree nodes within its half-height width. Genomes need >= 300 tree nodes with dS > 0 (5,978 genomes evaluated). A peak is called *sharp* if the half-height width is <= 1.2 natural-log units (a 3.3-fold range of dS), the peak holds >= 40% of nodes, and prominence is >= 0.8. Tiers: **A** sharp and BUSCO duplicated >= 40%; **B** sharp and 10-40%; **C** sharp and < 10%; **D** broad hump. BUSCO values are `duplicated_pct` from `tables/busco_genome.parquet` (fungi_odb12, joined by assembly ID). Peaks above 1.0 are not called, so hybrids whose second peak lies near 1.5 (for example the older Zygosaccharomyces parabailii peak) show only their lower peak.

974 of 5,978 genomes have an interior peak <= 1.0 (944 with one peak, 30 with two or more). Tier counts: A sharp + BUSCO dup>=40%: 83; B sharp + BUSCO dup 10-40%: 24; C sharp + BUSCO dup<10%: 28; D broad hump: 839.

Share of genomes with an interior peak by BUSCO duplication:

BUSCO dup bin	genomes	share_with_interior_peak
<1	4349	0.0963
1-5	962	0.263
5-10	227	0.238
10-20	162	0.364
20-40	115	0.539
>=40	163	0.779

Tier A by phylum and class:

phylum	cls	n_genomes_tier_A
Ascomycota	Saccharomycetes	23
Ascomycota	Pichiomycetes	17
Ascomycota	Sordariomycetes	13
Basidiomycota	Tremellomycetes	9
Ascomycota	Dothideomycetes	8
Basidiomycota	Agaricomycetes	5
Kickxellomycota	Dimargaritomycetes	2
Basidiomycota	Cystobasidiomycetes	2
Ascomycota	Ascomycota (unassigned)	1
Blastocladiomycota	Blastocladiomycetes	1
Entomophthoromycota	Entomophthoromycetes	1
Mucoromycota	Mucoromycetes	1

Known and literature hybrids in this screen (n_peaks = 0 means no interior peak <= 1.0, either no peak or a peak above 1.0):

genome	n_peaks	peak1_dS	peak1_mass	peak2_dS	peak2_mass	busco_dup	genes_ratio_clade
Aspergillus_latus_MO46169	0					77.5	1.76
Cryptococcus_neoformans_x_Cryptococcus_deneoformans_CBS_132	1	0.31	0.666			19	1.1
Ogataea_polymorpha_x_Ogataea_parapolymorpha_CBS_1977	1	0.167	0.551			78.6	2.92
Saccharomyces_pastorianus_CBS_1484	0					51.8	2
Verticillium_longisporum_PD589	1	0.158	0.714			77.2	1.54
Zygosaccharomyces_bailii_CLIB1307	1	0.167	0.779			64.1	1.97
Zygosaccharomyces_mellis_CBS_7431	1	0.162	0.748			64.3	1.89
Zygosaccharomyces_parabailii_PYCC5135	1	0.167	0.697			62.8	2.91
Zygosaccharomyces_pseudobailii	1	0.218	0.748			61.2	1.88
Zygosaccharomyces_rouxii_ATCC42981	1	0.178	0.75			64.2	2.06
Zygosaccharomyces_sapae_PYCC9313	1	0.154	0.789			64.8	2.89
Zygosaccharomyces_siamensis_PYCC9354	1	0.154	0.747			63.6	1.89

Shape-detected interior peaks. Orange: tier A, amber: tier B, blue: tier C, grey: broad. Dotted line: detected peak.

Peak position against BUSCO duplication, and the share of genomes with an interior peak by BUSCO duplication.



Figure 10: Shape-detected interior peaks. Orange: tier A, amber: tier B, blue: tier C, grey: broad. Dotted line: detected peak.

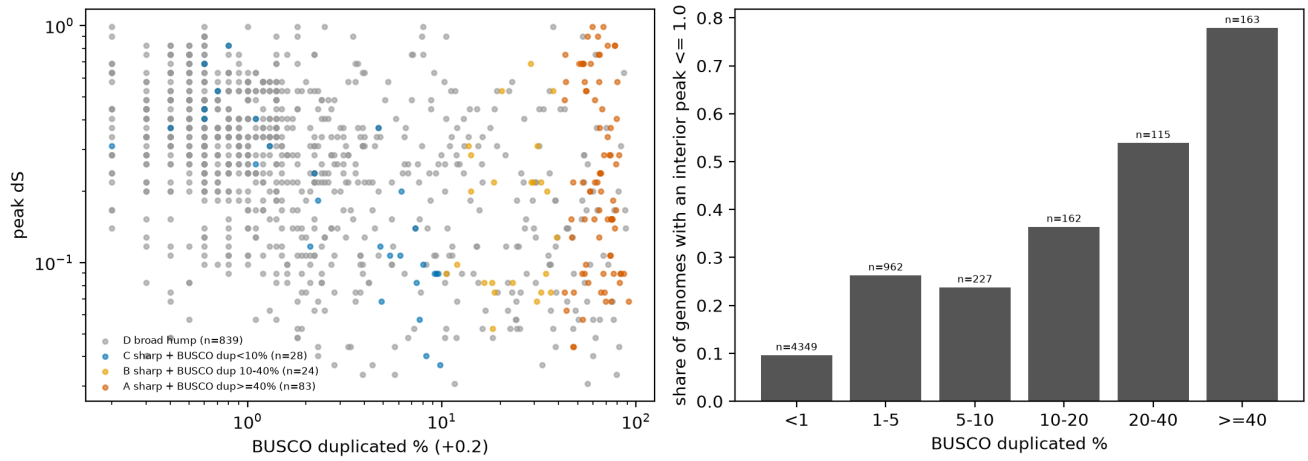


Figure 11: Peak position against BUSCO duplication, and the share of genomes with an interior peak by BUSCO duplication.

Reading the tiers. Tier A genomes combine a narrow dS peak with a high BUSCO duplicated fraction. The named and literature hybrids sit here (Verticillium longisporum, Ogataea polymorpha x parapolymorpha, several Zygosaccharomyces). This is consistent with an assembly that carries two diverged genomes, but dS and BUSCO together do not prove it. Gene count against the clade median, synteny and read data are the next checks (Section 10). Tier C genomes have a sharp peak but little BUSCO duplication. Some have gene counts far below the clade median (for example several Boletaceae with gene-count ratios of 0.1 to 0.5). That can mean partial annotation or a partial duplication, not a whole-genome event. Tier C is a list to inspect with FCS-GX and coverage data, not a list of events. The yeast peaks near dS 0.5 to 1.0 lie where yn00 reads low relative to codeml, so true parental divergence there is likely higher than the peak position.

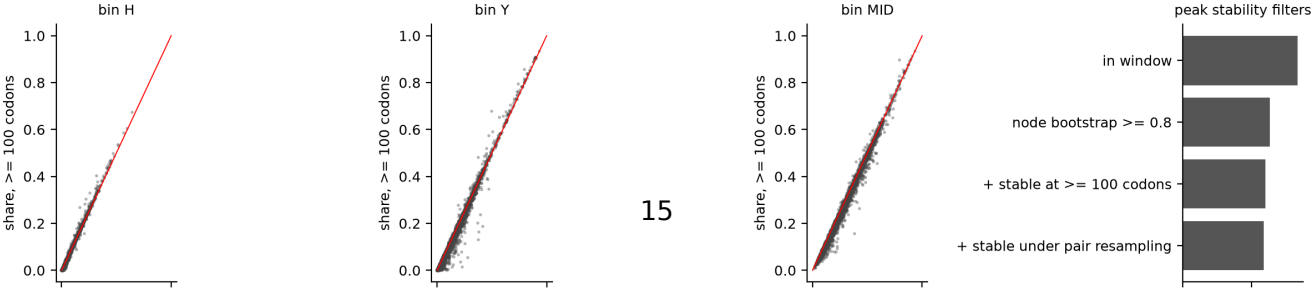
6. Assembly covariates and confounding

Fixed effects (contig count, size, GC, gene count, BUSCO duplication, TE per 1,000 genes) are in `tables/varcomp_fixed.tsv`. Compare the 'no covariates' and 'with covariates' rows in Section 3 to see how much clade variance is attributable to assembly properties. Where the clade share changes after adjustment, clade differences are not separable from assembly differences with these data.

7. Sensitivity to the codon gate (30 vs 100 codons)

bin	genomes	median_share_30codons	median_share_100codons	spearman	median_abs_diff
Z	7235	0.00222	0.00139	0.964	0.000279
H	7235	0.00885	0.00749	0.981	0.000825
Y	7235	0.0645	0.0569	0.991	0.00369
MID	7235	0.216	0.203	0.995	0.00978
OLD	7235	0.647	0.668	0.997	0.0121

spearman is the rank correlation of per-genome bin shares between the two gates.



yn00_dS_bin	n_nodes	median_yn00	median_codeml	median_ratio	ratio_p10	ratio_p90	share_within_20pct	pearson
(-0.001, 1e-09]	21088	0	0				0	
(1e-09, 0.02]	61297	0.00917	0.009	0.975	0.847	1.19	0.901	0.432
(0.02, 0.05]	69661	0.0347	0.0352	1.02	0.889	1.13	0.957	0.783
(0.05, 0.1]	108321	0.0743	0.0769	1.04	0.945	1.13	0.974	0.769
(0.1, 0.2]	171826	0.146	0.154	1.06	0.98	1.16	0.969	0.828
(0.2, 0.3]	124735	0.246	0.268	1.08	0.995	1.22	0.92	0.601
(0.3, 0.5]	165601	0.388	0.435	1.11	0.996	1.35	0.807	0.575
(0.5, 0.8]	143703	0.629	0.718	1.13	0.95	1.54	0.673	0.445
(0.8, 1.2]	111139	0.977	1.09	1.11	0.835	1.57	0.612	0.34
(1.2, 1.6]	89820	1.39	1.55	1.12	0.789	1.49	0.567	0.273
(1.6, 2.0]	52975	1.76	1.9	1.07	0.756	1.32	0.673	0.157
(2.0, 3.0]	14218	2.14	2.02	0.932	0.628	1.13	0.708	-0.0507

Genomes compared: 300. Median per-genome ratio (codeml / yn00): 1.009.

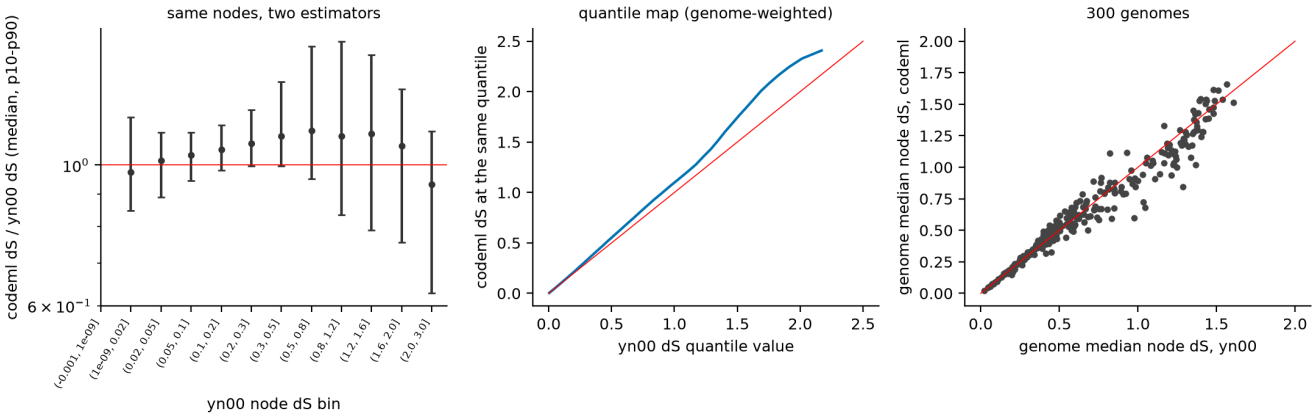


Figure 13: yn00 against codeml node dS.

yn00 against codeml node dS.

The ratio table shows the dS range where the two estimators agree and where they separate. Above about 0.5 the median codeml/yn00 ratio grows and the within-20% share falls. The 1.5 limit is stated as a measured property of this pipeline, not a general property of yn00. A codeml second pass for yn00 dS 0.5 to 1.6 is planned as a calibration subset.

9. Example genomes

Example genomes for identical-excess, H-excess and MID-excess flags.

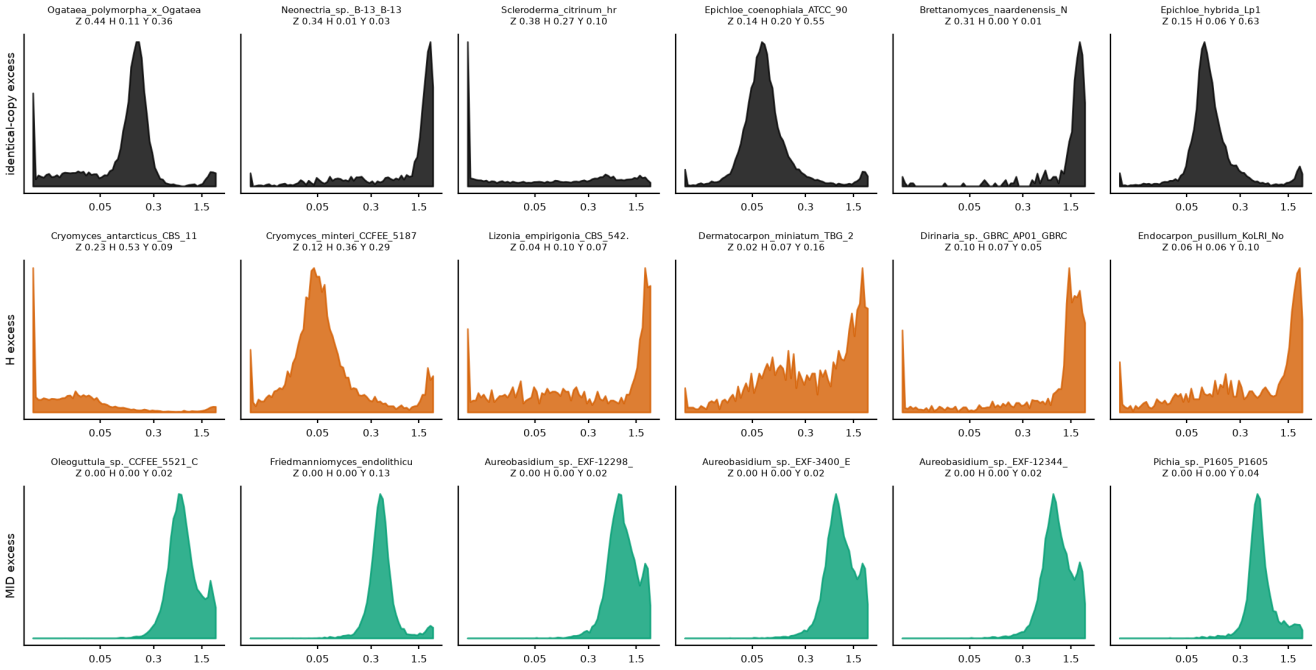


Figure 14: Example genomes for identical-excess, H-excess and MID-excess flags.

10. Follow-up tests for flagged genomes

1. BUSCO duplicated fraction and gene count against clade median (already in the tables).
2. Pfam domains of genes in identical groups (RVT_1, rve, DDE_Tnp, Chromo, gag). Report identical-group genes per 1,000 after removing TE-domain genes.
3. Contig-level checks: identical contigs, coverage bimodality, k-mer spectrum (heterozygous peak).
4. FCS-GX contig taxonomy for partial-genome peaks (possible contamination).
5. Synteny (wgd syn or i-ADHoRe) for any genome-scale claim. Do not use dS alone.
6. GFF coordinate overlap for annotation duplicates.

11. Literature context

Citations below were checked in PubMed on 2026-11-06 (see `fable_context.md` for DOIs and PMIDs).

- Yeast WGD: Wolfe and Shields 1997 (Nature); Kellis et al. 2004 (Nature); Marcet-Houben and Gabaldon 2015 (PLoS Biol), which proposes interspecies hybridisation as the origin.
- Mucorales: Ma et al. 2009 (PLoS Genet, Rhizopus delemar); Corrochano et al. 2016 (Curr Biol, Phycomyces and Mucor).
- Hybrids: Nakao et al. 2009 and Libkind et al. 2011 (S. pastorianus); Gallone et al. 2019 (beer hybrids); Ortiz-Merino et al. 2017 (Z. parabailii); Pryszcz et al. 2014 (C. orthopsilosis); Mixao and Gabaldon 2018 (review); Steenwyk et al. 2020 (A. latus); Fogelqvist et al. 2018 (V. longisporum); Lin et al. 2007 and Samarasinghe and Xu 2018 (Cryptococcus hybrids).
- Dikaryon and heterokaryon assemblies: Li et al. 2019 (Puccinia graminis haplotypes); Ropars et al. 2016, Chen et al. 2018 and Sperschneider et al. 2023 (AMF nuclei).
- TE-rich lineages: Duplessis et al. 2011 (rusts); Spanu et al. 2010 and Frantzeskakis et al. 2018 (Blumeria); Sipos et al. 2017 (Armillaria); Martin et al. 2010 (Tuber); Tisserant et al. 2013 (Rhizophagus); Goodwin et al. 2011 (Zymoseptoria).
- RIP: Galagan and Selker 2004 (Trends Genet).
- Assembly and Ks methods: Roach et al. 2018 (Purge Haplotigs); Steinegger and Salzberg 2020; Astashyn et al. 2024 (FCS-GX); Vanneste et al. 2013; Tiley et al. 2018; Zwaenepoel and Van de

Peer 2019 (two papers); Chen et al. 2024 (wgd v2); Yang and Nielsen 2000 (yn00).

Statements marked unverified in `fable_context.md` are not used here.

12. Gene-level queries: young paralogs and shared domains

Query design. Genes are those in the gated sampled pairs of `wgd.ks.parquet` (identical-CDS groups contribute one representative). Each gene is assigned the minimum dS over the sampled pairs it appears in, so a gene with several partners is placed in its youngest bin. Bins: Z (dS = 0), H1 (0, 0.1], Y (0.1, 0.3], MID (0.3, 1.5], OLD (> 1.5). *Recent* means Z plus H1. For each Pfam domain and each CAZy family, a Mantel-Haenszel odds ratio compares genes of the focal bin with all other paralog genes of the same genome and the same protein-length class (five classes: <150, 150-250, 250-400, 400-700, >=700 aa). Genome and length class are the strata, so large genomes and short proteins do not drive the result. For Pfam, only genes with at least one Pfam hit count in both groups, because the Pfam hit rate differs by bin (below). Variance is Robins-Breslow-Greenland; q is Benjamini-Hochberg across domains. A domain needs >= 20 genomes and >= 30 focal genes. Sets: all genomes, a *clean* set (BUSCO duplicated < 20%, gene count < 1.5x clade median, no named hybrid, not in shape tiers A or B), and clean Ascomycota and Basidiomycota separately. Two checks changed the first version of this test: strata with the domain only in the reference group must be kept, and the length strata were added after young genes proved shorter. Results from the earlier, unstratified version are kept in `tables/gene_enrich_unstratified/` for comparison.

Share of paralog genes with at least one Pfam hit, by bin (all genomes):

bin	n	frac_with_pfam
OLD	6218262	0.831
H1	3428502	0.511
MID	5403758	0.532
Y	3513173	0.491
Z	869298	0.459

Protein features of each bin against OLD (> 1.5) in the same genome, clean genomes, paired across genomes with >= 10 genes in both bins (Wilcoxon):

bin	metric	n_genomes	median_B	median_OLD	median_diff	frac_genomes_higher	p_wilcoxon
Z	med_len	2597	260	428	-158	0.0774	0
Z	frac_short150	2597	0.252	0.0569	0.184	0.933	0
Z	frac_sigp	2597	0.0328	0.109	-0.0667	0.17	2.76e-232
H1	med_len	4452	291	431	-129	0.122	0
H1	frac_short150	4452	0.2	0.0549	0.13	0.917	0
H1	frac_sigp	4452	0.0336	0.105	-0.0639	0.166	0
Y	med_len	5188	291	433	-137	0.0715	0
Y	frac_short150	5188	0.215	0.0532	0.151	0.954	0
Y	frac_sigp	5188	0.0485	0.104	-0.048	0.227	0
MID	med_len	6888	320	434	-111	0.034	0
MID	frac_short150	6888	0.177	0.051	0.12	0.983	0
MID	frac_sigp	6888	0.133	0.108	0.0218	0.64	1.49e-235

Young paralog genes are shorter and less often carry a signal peptide than OLD paralogs, in most genomes. Part of the length difference can arise from detection: a short protein is less likely to keep an alignable, gated pair at high dS. The enrichment test therefore stratifies on length.

Number of domains enriched (q < 0.01, OR >= 2) by bin and set:

bin	set	kind	genomes	domains_tested	enriched_q01_OR2
recent (dS <= 0.1)	all	Pfam	4195	5623	854
recent (dS <= 0.1)	all	CAZy	5233	146	2
recent (dS <= 0.1)	clean	Pfam	3768	4381	1227
recent (dS <= 0.1)	clean	CAZy	4788	112	3
recent (dS <= 0.1)	clean_Ascomycota	Pfam	2150	1389	456
recent (dS <= 0.1)	clean_Ascomycota	CAZy	3013	57	2
recent (dS <= 0.1)	clean_Basidiomycota	Pfam	1205	2755	431
recent (dS <= 0.1)	clean_Basidiomycota	CAZy	1304	80	1
identical (dS = 0)	all	Pfam	2162	3185	268
identical (dS = 0)	all	CAZy	3084	85	2
identical (dS = 0)	clean	Pfam	1834	1256	267
identical (dS = 0)	clean	CAZy	2706	51	1
identical (dS = 0)	clean_Ascomycota	Pfam	959	559	126
identical (dS = 0)	clean_Ascomycota	CAZy	1449	27	2
identical (dS = 0)	clean_Basidiomycota	Pfam	646	375	54
identical (dS = 0)	clean_Basidiomycota	CAZy	941	11	0
Y (0.1-0.3)	all	Pfam	4624	5483	165
Y (0.1-0.3)	all	CAZy	5582	143	1
Y (0.1-0.3)	clean	Pfam	4192	3649	290
Y (0.1-0.3)	clean	CAZy	5127	111	2
Y (0.1-0.3)	clean_Ascomycota	Pfam	2520	887	233
Y (0.1-0.3)	clean_Ascomycota	CAZy	3349	37	2
Y (0.1-0.3)	clean_Basidiomycota	Pfam	1239	2757	128
Y (0.1-0.3)	clean_Basidiomycota	CAZy	1305	85	0
MID (0.3-1.5)	all	Pfam	7274	6158	311
MID (0.3-1.5)	all	CAZy	7361	160	4
MID (0.3-1.5)	clean	Pfam	6814	4751	418
MID (0.3-1.5)	clean	CAZy	6899	149	3
MID (0.3-1.5)	clean_Ascomycota	Pfam	4936	2478	495
MID (0.3-1.5)	clean_Ascomycota	CAZy	4969	106	8
MID (0.3-1.5)	clean_Basidiomycota	Pfam	1433	3271	115
MID (0.3-1.5)	clean_Basidiomycota	CAZy	1452	112	1

Top Pfam domains for recent paralogs, clean genomes (>= 300 focal genes, sorted by odds ratio):

dom	name	n_genomes	focal_genes	OR	ci_lo	ci_hi
PF:PF27657	RRT15	138	481	22.7	17.1	30.3
PF:PF07282	Cas12f1-like_TNB	123	1.81e+03	14.3	13.1	15.7
PF:PF12511	DUF3716	214	902	12.1	10.7	13.8
PF:PF26935	DUF8286	53	553	8.96	7.83	10.3
PF:PF28402	GloB_C	124	364	8.25	7.02	9.7
PF:PF12520	DUF3723	266	964	7.75	7	8.58
PF:PF13837	Myb_DNA-bind_4	214	966	6.8	6.08	7.61
PF:PF12013	OrsD	532	2.54e+03	6.79	6.37	7.25
PF:PF23042	KOW1_SPT5	246	1.56e+03	6.43	5.84	7.08
PF:PF12776	Myb_DNA-bind_3	429	2.62e+03	6.43	6	6.89
PF:PF13966	zf-RVT	90	534	6.37	5.61	7.24
PF:PF03439	Spt5-NGN	185	1.12e+03	6.32	5.67	7.06
PF:PF29418	DUF3723_N	297	910	6.27	5.68	6.91
PF:PF12851	Tet_JBP	231	1.41e+03	6.25	5.71	6.84
PF:PF18721	CxC6	204	1.73e+03	5.31	4.89	5.76
PF:PF18718	CxC5	227	1.96e+03	5.04	4.68	5.44
PF:PF20499	DUF6729	122	549	4.89	4.27	5.6

dom	name	n_genomes	focal_genes	OR	ci_lo	ci_hi
PF:PF13352	DUF4100	269	1.43e+03	4.8	4.41	5.22
PF:PF24764	rva_4	492	2.62e+03	4.75	4.48	5.05
PF:PF27653	SPOK2_N	270	1.21e+03	4.48	4.13	4.87

Top Pfam domains for recent paralogs, clean Ascomycota:

dom	name	n_genomes	focal_genes	OR	ci_lo	ci_hi
PF:PF24764	rva_4	90	413	16.1	12.9	20
PF:PF20149	DUF6532	81	1.61e+03	15.3	13.5	17.2
PF:PF18759	Plavaka	88	2.87e+03	14	12.8	15.2
PF:PF12511	DUF3716	214	902	12.1	10.7	13.8
PF:PF12013	OrsD	287	1.71e+03	11.2	10.2	12.2
PF:PF20231	DUF6589	67	337	9.85	7.93	12.2
PF:PF29418	DUF3723_N	245	825	8.58	7.67	9.59
PF:PF28402	GloB_C	120	359	8.16	6.93	9.6
PF:PF12520	DUF3723	264	959	7.72	6.98	8.55
PF:PF02902	Peptidase_C48	135	322	5.91	4.98	7.03
PF:PF05699	Dimer_Tnp_hAT	158	738	5.45	4.83	6.15
PF:PF22594	GTP-eEF1A_C	208	441	5.11	4.42	5.9
PF:PF03143	GTP_EFTU_D3	218	457	4.89	4.25	5.63
PF:PF27653	SPOK2_N	270	1.21e+03	4.48	4.13	4.87
PF:PF11917	DUF3435	400	1.62e+03	4.24	3.96	4.54

Top Pfam domains for recent paralogs, clean Basidiomycota:

dom	name	n_genomes	focal_genes	OR	ci_lo	ci_hi
PF:PF13837	Myb_DNA-bind_4	171	629	7.24	6.24	8.4
PF:PF12776	Myb_DNA-bind_3	365	2.28e+03	6.66	6.17	7.18
PF:PF12851	Tet_JBP	212	1.35e+03	6.48	5.9	7.11
PF:PF23042	KOW1_SPT5	219	1.51e+03	6.44	5.85	7.09
PF:PF03439	Spt5-NGN	157	1.06e+03	6.32	5.66	7.07
PF:PF18721	CxC6	203	1.73e+03	5.31	4.89	5.77
PF:PF18718	CxC5	227	1.96e+03	5.04	4.68	5.44
PF:PF02338	OTU	54	339	4.96	4.08	6.03
PF:PF13352	DUF4100	266	1.42e+03	4.77	4.39	5.19
PF:PF18803	CxC2	492	6.66e+03	4.32	4.15	4.49
PF:PF28437	DUF8397	106	587	4.29	3.76	4.89
PF:PF26792	DUF8263	176	1.01e+03	4.28	3.88	4.71
PF:PF10551	MULE	116	323	4.27	3.6	5.06
PF:PF20681	DUF6818	125	370	4.22	3.6	4.94
PF:PF24764	rva_4	393	2.17e+03	4.15	3.89	4.42

Top Pfam domains for identical-CDS genes, clean genomes:

dom	name	n_genomes	focal_genes	OR	ci_lo	ci_hi
PF:PF27657	RRT15	92	314	22.2	17.3	28.6
PF:PF12511	DUF3716	85	358	5.84	5.01	6.82
PF:PF12013	OrsD	165	711	5.3	4.81	5.83
PF:PF18718	CxC5	74	466	4.1	3.67	4.57

dom	name	n_genomes	focal_genes	OR	ci_lo	ci_hi
PF:PF18721	CxC6	66	414	4	3.56	4.5
PF:PF12851	Tet_JBP	70	276	3.87	3.37	4.45
PF:PF24764	rva_4	166	572	3.78	3.44	4.16
PF:PF01693	Cauli_VI	128	527	3.51	3.18	3.88
PF:PF23042	KOW1_SPT5	69	271	3.4	2.95	3.9
PF:PF20414	DUF6698	200	1.32e+03	3.2	3	3.4
PF:PF01399	PCI	121	284	3.05	2.61	3.56
PF:PF02902	Peptidase_C48	131	446	2.97	2.66	3.31
PF:PF00867	XPG_I	98	269	2.88	2.51	3.3
PF:PF18803	CxC2	187	1.05e+03	2.7	2.52	2.89
PF:PF20231	DUF6589	149	558	2.47	2.26	2.71

Top Pfam domains for Y-bin genes (0.1-0.3), clean genomes:

dom	name	n_genomes	focal_genes	OR	ci_lo	ci_hi
PF:PF08071	RS4NT	235	432	5.75	5.07	6.51
PF:PF00181	Ribosomal_L2_N	335	638	5.69	5.1	6.34
PF:PF03947	Ribosomal_L2_C	322	622	5.58	5	6.22
PF:PF16121	40S_S4_C	237	437	5.48	4.84	6.2
PF:PF00900	Ribosomal_S4e	235	429	5.44	4.8	6.16
PF:PF11917	DUF3435	535	1.62e+03	5.22	4.9	5.56
PF:PF26935	DUF8286	74	455	5.05	4.42	5.77
PF:PF23948	ARM_5	48	1.34e+03	5.03	4.63	5.46
PF:PF12520	DUF3723	236	553	4.77	4.31	5.28
PF:PF00805	Pentapeptide	56	1.24e+03	4.67	4.28	5.09
PF:PF01015	Ribosomal_S3Ae	149	334	4.18	3.67	4.77
PF:PF01479	S4	369	722	4.01	3.65	4.41
PF:PF15715	PAF	361	718	3.92	3.53	4.37
PF:PF01157	Ribosomal_L21e	167	310	3.9	3.36	4.53
PF:PF00416	Ribosomal_S13	155	314	3.84	3.3	4.46

CAZy families enriched among recent paralogs, clean genomes (>= 30 focal genes):

dom	name	n_genomes	focal_genes	OR	ci_lo	ci_hi
CAZy:GT59		22	38	2.95	1.72	5.07
CAZy:GT33		29	51	2.32	1.57	3.41
CAZy:GT76		38	78	2.28	1.63	3.18

Pfam enrichment volcano plots (clean genomes) and protein length by dS bin.

Interpretation, with limits.

- Domain names are those in the Pfam table. Many top domains are named for transposases, integrases or retroelement proteins (for example Plavaka, KDZ, CxC, Tet_JBP, MULE, Transposase_21, rva_4, zf-RVT, Myb_DNA-bind_3 and _4, Cas12f1-like_TNB). I did not check each family description here. By name, this set points to transposon-derived gene models among recent duplicates. This matches the TE-like identical-copy profile in Section 4.
- SPOK2_N and OrsD (PF12013) are enriched among recent paralogs, mainly in Ascomycota. I did not verify what these families do. The OrsD gene set is examined in `reports/H3_OrsD_case_study.md`.

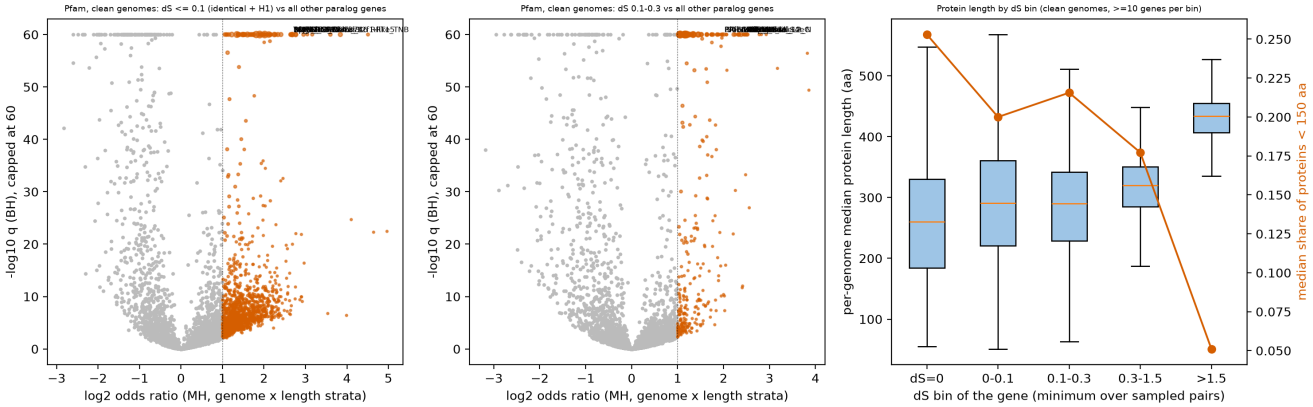


Figure 15: Pfam enrichment volcano plots (clean genomes) and protein length by dS bin.

- Y-bin genes are enriched for ribosomal-protein and translation-factor domains (L2, S4, S6e, L21e). Ribosomal-protein families are small and often multicopy, so duplicated models or processed copies are plausible. This is not tested here.
- Several domains are of unknown function (DUF names, RRT15). They are reported as found and not interpreted.
- CAZy families show almost no enrichment in the recent bin after stratification. Recent duplication of CAZymes is not a general feature of these data.
- The dS bin is set by sampled pairs, not by all pairs of a gene, and each family contributes at most five pairs per node. Enrichment of a domain may partly reflect how families are sampled. The effect of this sampling on gene-level bins has not been measured.

13. Reproducibility

Scripts: `eval/explore/scripts/` (shape_peaks, shape_plots, gene_recent_enrich, gene_plots, gene_stats, taxonomy, build_features2, build_table, pfam_te_counts, build_hist, compare_codeml_yn00, stats, varcomp.R, plots, make_report). Inputs: `paralogoscope_pilot/nf_full/tables_yn00` (interim, 7,384 genomes; to be replaced by the final tables when the last giant genome completes). Report built 2026-11-07 09:55.